DO NOW — RECALL

Paper 2 targeted reteaching

Name: _____ Class: _____ Date: _____

1. What is O(log n)?

2. What is a recursive base case?

3. What does a trace table show?

4. Trace search for 3 in sorted [1,3,5,7,9].

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Work growing with the number of halvings.

2. Model answer

Condition stopping further calls.

3. Model answer

Intermediate program states.

4. Model answer

mid2→5, high1, mid0→1, low1, mid1→3; return index1.

By the end, you can…

• Correct binary-search bound updates

• Repair recursive termination

• Justify algorithm complexity with evidence

Vocabulary — know these terms

bounds — Limits on a search interval.

base case — Recursive stopping condition.

auxiliary space — Extra storage beyond input.

NEW LEARNING

Read, annotate and explain

Identify

Locate incorrect condition or state update

Model

Model a correct trace table

Apply

Test new data, including missing target

Evaluate

Explain time and memory consequences

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace search for 3 in [1,3,5,7,9].

Explain the method

The midpoint is excluded after a failed comparison, so bounds shrink.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace binary search for 4 in [2,4,6,8,10], giving bounds/midpoints. [6]

Question 2

Trace recursive SumTo(4), including base0 and return values. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare bubble and merge sort worst-case time/extra space; choose for million values with sufficient memory. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Binary search may never update the bounds but still always terminate.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.